11. $f(x) = \int_x^0 \sqrt{1+\sec t}\, dt$

$= -\frac{d}{dx}\int_0^x \sqrt{1+\sec t}\, dt$

$= \sqrt{1+\sec t}\, dt$

13. $h(x) = \int_1^{e^x} \ln t\, dt$

let $u = e^x$, $\frac{du}{dx} = e^x$

$\frac{d}{dx} = \frac{du}{dx} \cdot \frac{d}{du}$

$= \frac{d}{dx}\int_1^u \ln u\, dt$ $= \ln(e^x) \cdot \frac{d}{dx}(e^x)$ (chain rule)

$= \frac{du}{dx}\left[\frac{d}{du}\cdot\int_1^u \ln t\, dt\right]$ → $[\ln t]_1^u$

$= e^x \cdot \ln e^x = x \cdot e^x$

14. $h(x) = \int_1^{\sqrt{x}} \frac{z^2}{z^4+1}\,dz$

$\frac{d}{dx}\int_1^{\sqrt{x}} \frac{z^2}{z^4+1}\,dz$

$= \frac{z^2}{z^4+1} \cdot \frac{1}{2\sqrt{x}}$

$\frac{\sqrt{x}^2}{\sqrt{x}^4+1}$

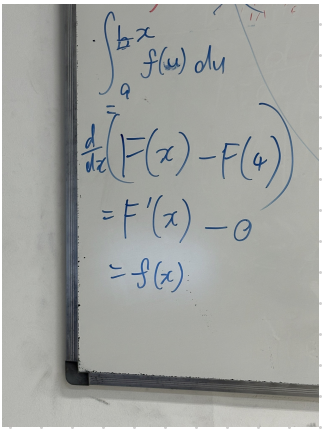
$u = \sqrt{x}$

$\frac{du}{dx} \cdot \frac{d}{du}\int_1^{u} \frac{z^2}{z^4+1}\,dz$

u 代入

$\frac{u^2}{u^4+1} = \frac{\sqrt{x}^2}{\sqrt{x}^4+1}$

$F(u) - F(1)$



28. $\int_0^1 (u+2)(u+3)\,du$

$= \int_0^1 u^2+5u+6\,du$

$= \left[\frac{u^3}{3} + \frac{5u^2}{2} + 6u\right]_0^1$

$= \frac{1}{3} + \frac{5}{2} + 6 = \frac{17}{6} + 6$

31. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc t \cot t\,dt$

$= \left[-\csc t\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$

$= \left[-\frac{1}{\sin\frac{\pi}{2}} + \frac{1}{\sin(\frac{\pi}{6})}\right]$

$= 1$

55. $\int_1^2 \frac{v^3+5v^6}{v^4}\,dv$

$= \int_1^2 \frac{1}{v} + 5v^2\,dv$

$= \left[\ln|v| + \frac{5v^3}{3}\right]_1^2$

$= \left\{\ln|v| + v^3 - \ln|1| - 1\right] = 7.693$